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Quiz 5

NAME: ____

Q.1. Write the general equation of the circle with center $C(h, k)$ and radius r. The following is the equation of a circle in the $x - y$ plane: $x^2 - 4x + y^2 = 5$. With working, find its center and radius (**Hint:** start by completing the squares for the x-terms and y-terms separately).

Let's group the x-terms and y-terms together and complete the squares:

$$(x^2 - 4x) + y^2 = 5$$

Notice that y^2 is already a completed square! $y^2 = (y - 0)^2$.

For $x^2 - 4x$, following our usual procedure, we factor out the coefficient of x^2 which is 1 (not really changing the expression in this case) and then divide the coefficient of x by 2. This gives us $\frac{-4}{2} = -2$. So our guess for the completed square of just the x-terms is:

$$x^2 - 4x = (x - 2)^2 + d$$

and we just need to solve for d. Expanding the right side, we get:

$$\begin{aligned} \cancel{x^2} - \cancel{4x} &= \cancel{x^2} - \cancel{4x} + 4 + d \\ \Rightarrow 0 &= 4 + d \\ \Rightarrow d &= -4 \end{aligned}$$

So our completed square form for $x^2 - 4x$ is $(x - 2)^2 - 4$. Plugging this back into the circle equation, we get:

$$\begin{aligned} (x-2)^2 - 4 + y^2 &= 5 \\ \Rightarrow (x-2)^2 + y^2 &= 9 \\ \Rightarrow (x-2)^2 + (y-0)^2 &= 3^2 \end{aligned}$$

so our circle center is $\boxed{(2, 0)}$ and our circle radius is $\boxed{3}$.

Q.2. I have a line l in the $x - y$ plane. l is perpendicular to the line $y = 1 - x$ and passes through the point $(-1, 4)$. What is the **slope-intercept** form of l? (**Hint:** start by setting up the point-slope form of l)

The general point-slope form is $y - y_1 = m(x - x_1)$ where m is the slope of the line and (x_1, y_1) is a point on the line. We know that $(x_1, y_1) = (-1, 4)$ is a point on the line, so our point-slope form becomes:

$$\begin{aligned} y - 4 &= m(x - (-1)) \\ \Rightarrow y - 4 &= m(x + 1) \end{aligned}$$

We now just need to find m. We know our new line is perpendicular to the line $y = 1 - x$ which has slope -1. We know two lines with slopes m_1, m_2 are perpendicular exactly when $m_2 = -\frac{1}{m_1}$ (or vice versa). So the slope of our new line has to be $\frac{-1}{-1} = 1$. Plugging this into the point-slope formula:

$$\begin{aligned} y - 4 &= 1 \cdot (x+1) \\ \Rightarrow y - 4 &= x + 1 \end{aligned}$$

We are not done yet! The question asked for the slope-intercept form of this line, so we need to rearrange the equation to isolate y on one side:

$$\begin{aligned} y - 4 &= x + 1 \\ \Rightarrow y &= x + 5 \end{aligned}$$

so the slope-intercept form of our new line is $\boxed{y = x + 5}$.

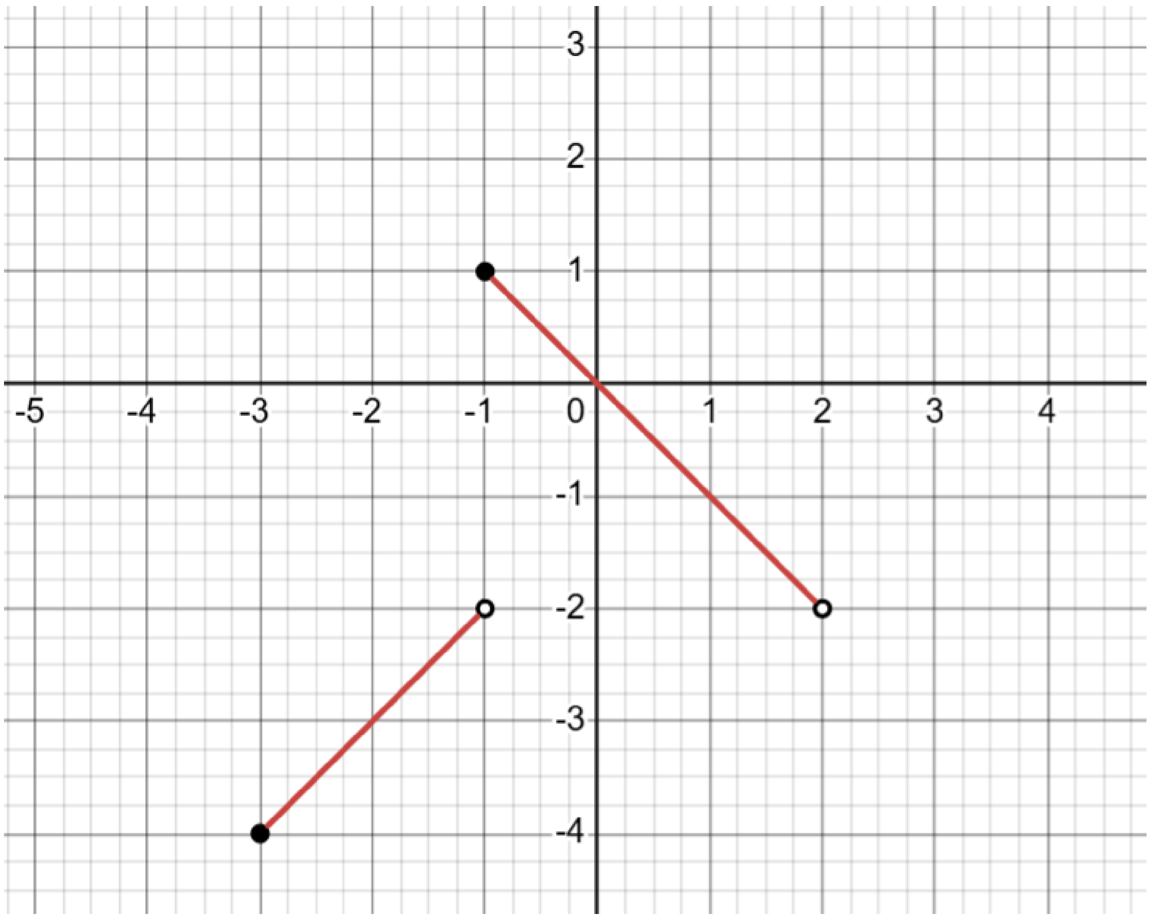
Q.3. Find the domain of the function $f(x) = \frac{\sqrt{x+1}}{2x}$. Explain all restrictions on the domain clearly (either with equations/ inequalities or in words). Write your final answer in interval notation.

The restrictions on the domain for this function will arise from the fact that its definition involves a square-root and a division. We cannot square root negative numbers (for real-valued functions) and we cannot divide by 0. So what does that mean for our domain?

Well for x to be in the domain, we better have $2x \neq 0 \Rightarrow x \neq 0$, otherwise we get a 0 in the denominator. But **also** for x to be in the domain, we better have $x + 1 \geq 0$, otherwise $x + 1$ will be negative and there will be a negative number under the square root.

$x + 1 \geq 0$ means $x \geq -1$. Written via interval notation, this condition means that x has to be in the interval $[-1, \infty)$. But we can't forget our earlier restriction. We must **also** have $x \neq 0$. Since x must lie in the interval $[-1, \infty)$ and also x must not be 0, our final answer for the full domain written in interval notation is $\boxed{[-1, 0) \cup (0, \infty)}$.

Q.4. The graph of a function $y = f(x)$ is given below:



State the domain and range of $f(x)$ in interval notation. In interval notation, state the intervals on which $f(x)$ is increasing and decreasing, respectively. Clearly mark all your answers.

The domain is the horizontal spread of the graph. Solid points are explicitly included as parts of the graph, but hollow points are explicitly excluded as parts of the graph. For the domain, we see a horizontal spread of $[-3,-1)$ for the left line of the graph and a horizontal spread of $[-1,2)$ for the right line of the graph. So our total graph horizontal spread is $[-3,-1) \cup [-1,2) = \boxed{[-3,2)}$.

For the range, we see a vertical spread of $[-4,-2)$ for the left line of the graph and a vertical spread of $(-2,1)$ for the right line of the graph. So our total graph vertical spread is $\boxed{[-4,-2) \cup (-2,1)}$ (and this cannot be simplified into a single interval).

We see that the graph is increasing from $x = -3$ all the way to $x = -1$, and yes that includes $x = -1$. The value $f(-1)$ is 1 which is still bigger than all the values of $f(x)$ when $x < -1$ (and increasing just means that bigger inputs should always give bigger outputs which is true here even including $x = -1$). So the graph is increasing on the interval $\boxed{[-3,-1]}$. The graph is then decreasing from $x = -1$ up to $x = 2$ but not including $x = 2$ itself ($x = 2$ is not actually in our domain). So the graph is decreasing on the interval $\boxed{[-1, 2)}$.